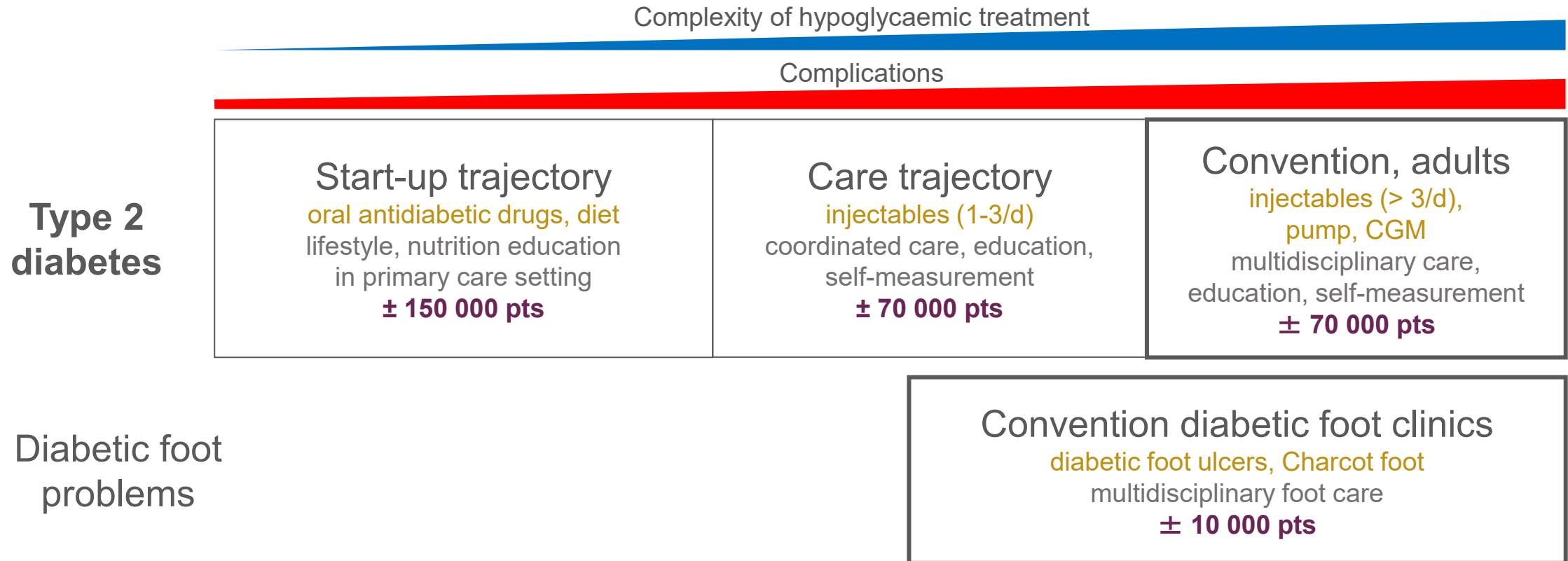


THE USE OF DATA TO IMPLEMENT AND EVALUATE DIABETES POLICY: PERSPECTIVE FROM THE BELGIAN PUBLIC HEALTH RESEARCH INSTITUTE

Astrid Lavens

EASD-EUDF Symposium2025, Vienna, September 17

Organisation of Belgian diabetes care



Organisation of Belgian diabetes care

Diabetic foot
problems

Convention diabetic foot clinics
diabetic foot ulcers, Charcot foot
multidisciplinary foot care
± 10 000 pts

Complexity of hypoglycaemic treatment



Complications

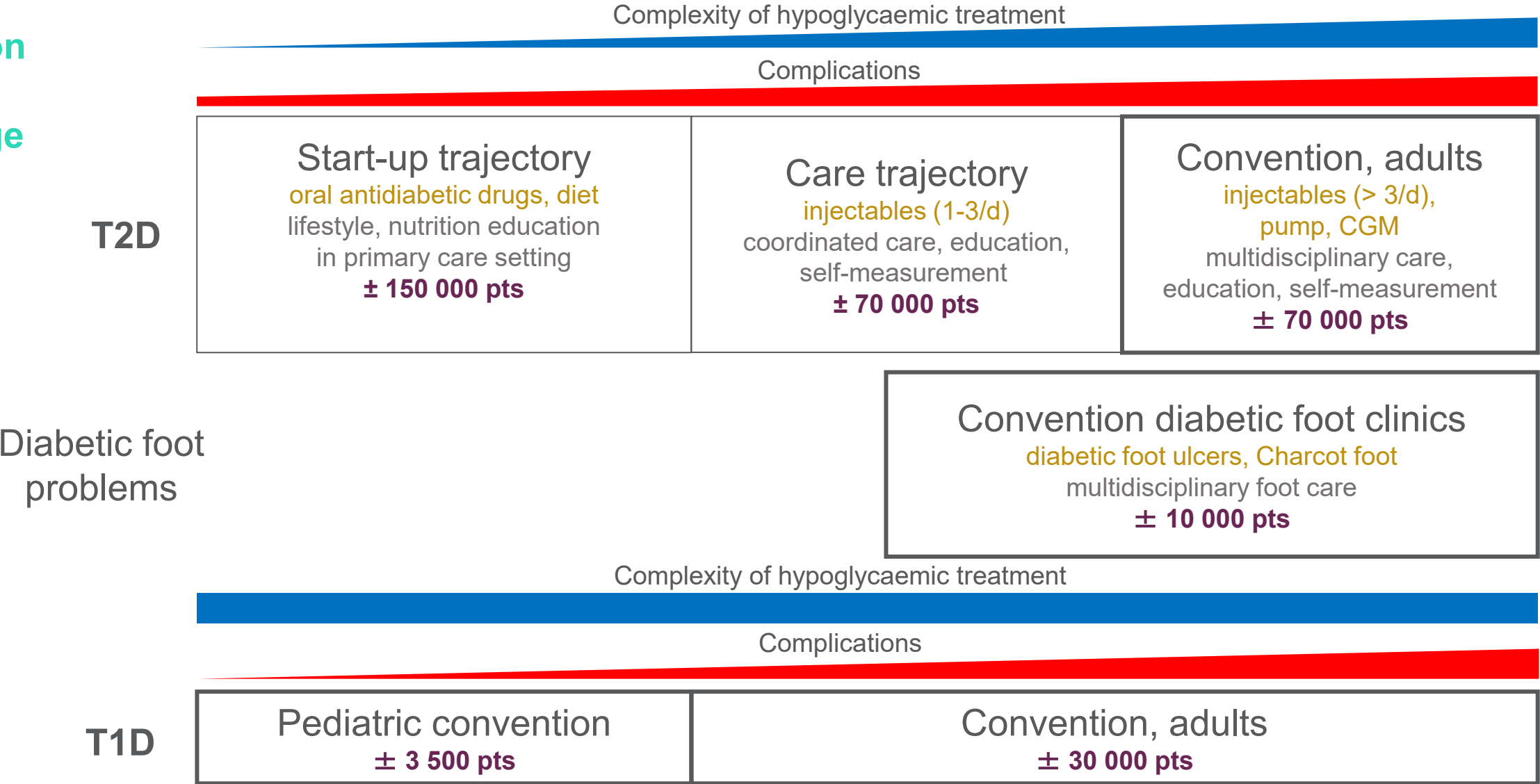


Type 1
diabetes

Pediatric convention ± 3 500 pts	Convention, adults ± 30 000 pts
-------------------------------------	------------------------------------

Organisation of Belgian diabetes care

Data
collection
= Big
challenge



Organisation of Belgian diabetes care (2nd and 3th line) : The diabetes conventions

Diabetes convention

= National care program promoting self-regulation for people with diabetes (since 1988 for adults, since 1997 for children and adolescents)



Diabetes centers

(* Endocrinologist or pediatrician, diabetes educators and nurses, dietitians, podiatrists and psychologists)

= healthcare financing and health insurance body

Free of charge:

- Follow up by multidisciplinary team *
- Diabetes education
- Material for diabetes self management (CGM, CSII, HCL-systems)

People with diabetes



Organisation of Belgian diabetes care (2nd and 3th line) : The diabetes conventions

Diabetes convention

= National care program promoting self-regulation for people with diabetes (since 1988 for adults, since 1997 for children and adolescents)



Diabetes centers

(Endocrinologist or pediatrician, Diabetes educator and nurses, dietitians, podiatrists and psychologists)

= healthcare financing and health insurance body

Free of charge:

- Follow up by multidisciplinary team
- Diabetes education
- Material for diabetes self management (CGM, CSII, HCL-systems)

People with diabetes



- **Nearly all people with T1D and T2D on intensive insulin participate in this program**

Organisation of Belgian diabetes care (2nd and 3th line) : The diabetes conventions

Diabetes convention

= National care program promoting self-regulation for people with diabetes (since 1988 for adults, since 1997 for children and adolescents)



Diabetes centers

(Endocrinologist or pediatrician, Diabetes educator and nurses, dietitians, podiatrists and psychologists)

= healthcare financing and health insurance body

Free of charge:

- Follow up by multidisciplinary team
- Diabetes education
- Material for diabetes self management (CGM, CSII, HCL-systems)

People with diabetes



➤ **Obligatory** participation to quality improvement initiatives

Organisation of Belgian diabetes care (2nd and 3th line) : The diabetes conventions



NIHDI

Obligatory
participation
to quality
improvement
initiatives

T2D

Complexity of hypoglycaemic treatment

Complications

Start-up trajectory

oral antidiabetic drugs, diet
lifestyle, nutrition education
in primary care setting

Care trajectory

injectables (1-3/d)
coordinated care, education,
self-measurement

Convention, adults

injectables (> 3/d),
pump, CGM
multidisciplinary care,
education, self-measurement

IQED (2001)



Diabetic foot
problems

Convention diabetic foot clinics

diabetic foot ulcers, Charcot foot
multidisciplinary foot care

IQED-Foot (2005)

Sciensano
3 diabetes
projects

T1D

Complexity of hypoglycaemic treatment

Complications

Pediatric convention

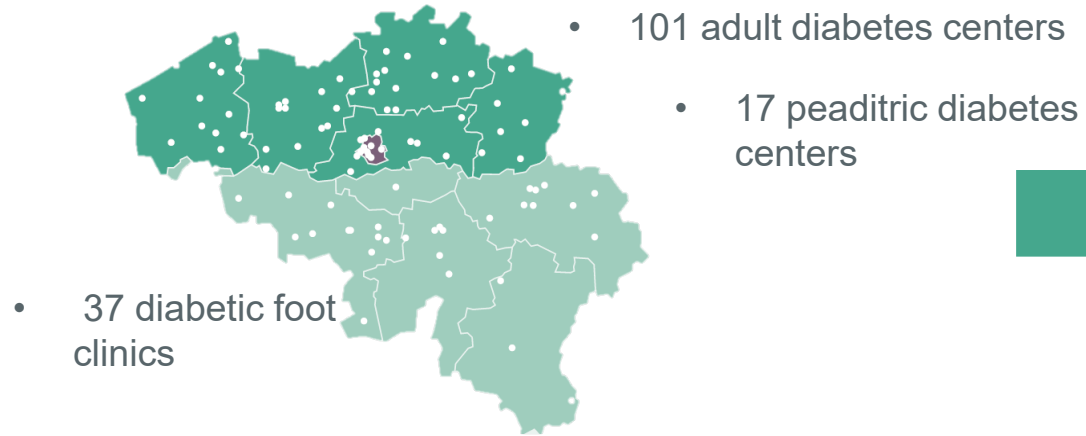
IQECAD (2008)

Convention, adults

IQED (2001)

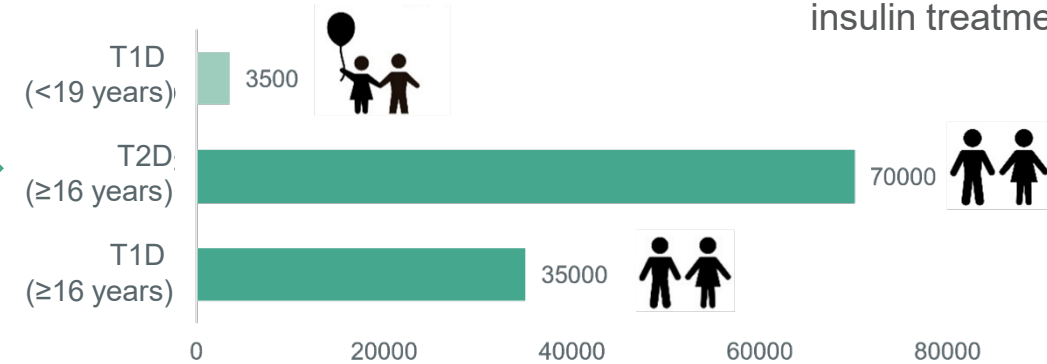
Initiative for Quality Improvement and Epidemiology in Diabetes (IQED)

Hospital based diabetes centers



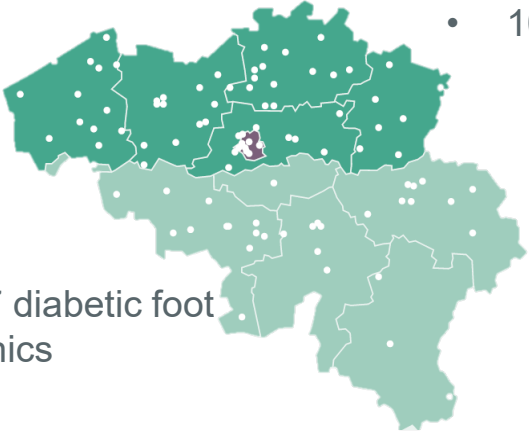
Who?

People followed in convention = people with T1D and T2D on intensive insulin treatment



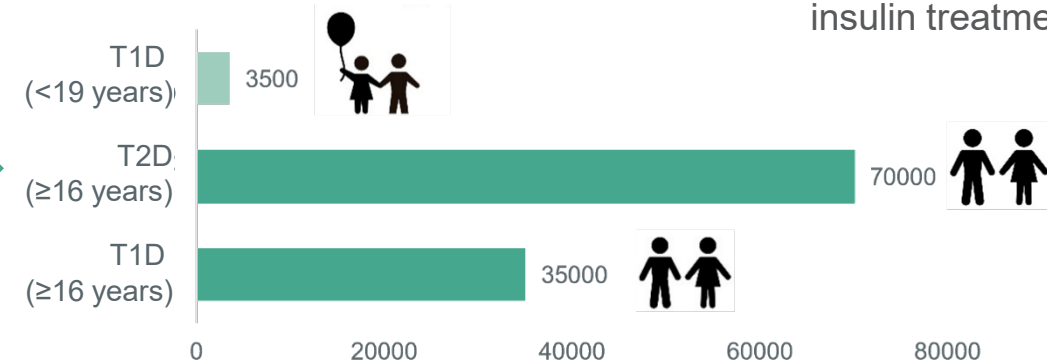
Initiative for Quality Improvement and Epidemiology in Diabetes (IQED)

Hospital based diabetes centers

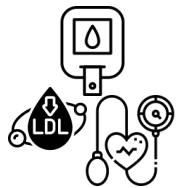
- 101 adult diabetes centers
 - 17 pediatric diabetes centers
 - 37 diabetic foot clinics
- 

Who?

People followed in convention = people with T1D and T2D on intensive insulin treatment



Data? (2-yearly)



Risk factors



Metabolic health



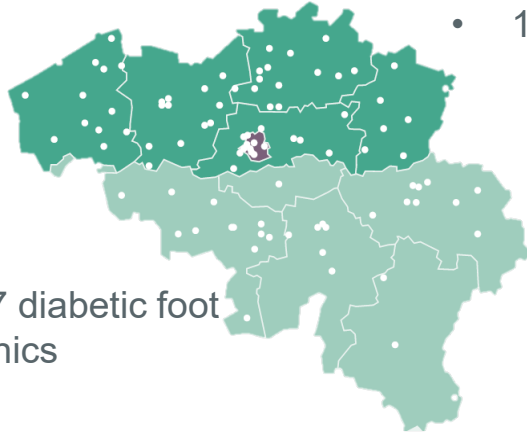
Complications



Treatment

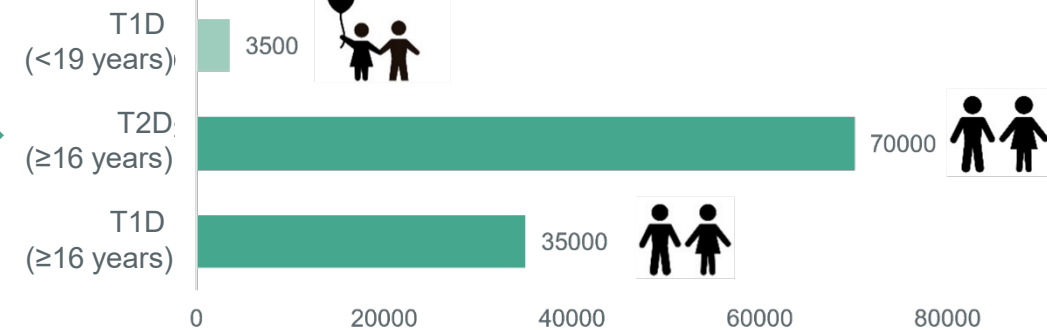
Initiative for Quality Improvement and Epidemiology in Diabetes (IQED)

Hospital based diabetes centers

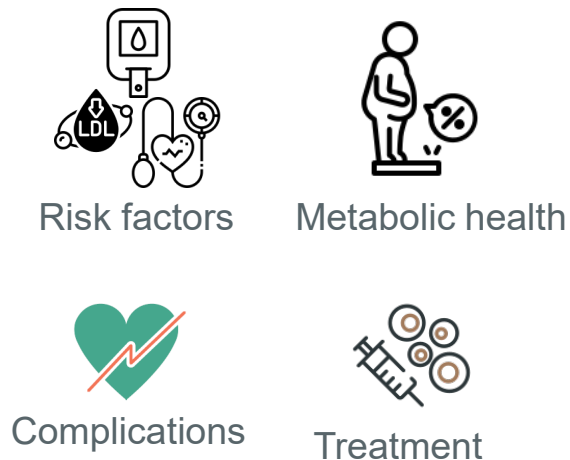
- 101 adult diabetes centers
 - 17 pediatric diabetes centers
 - 37 diabetic foot clinics
- 

Who?

People followed in convention = people with T1D and T2D on intensive insulin treatment

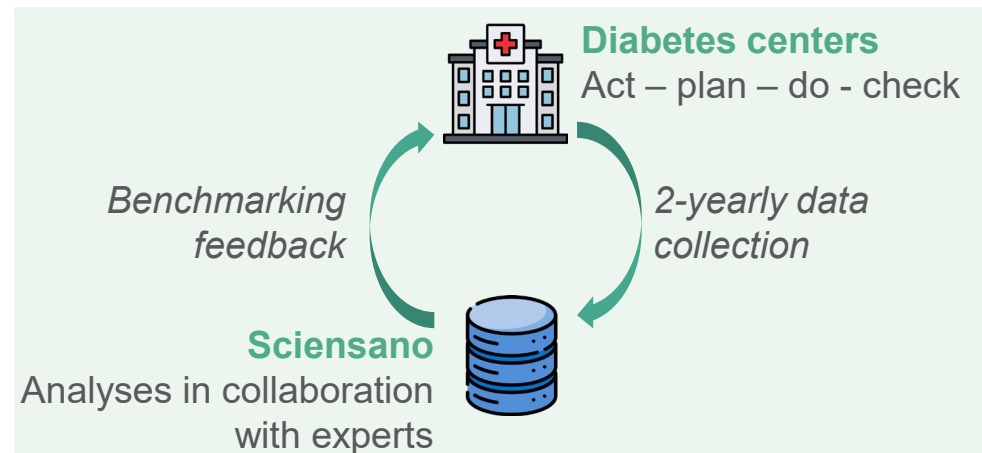


Data? (2-yearly)



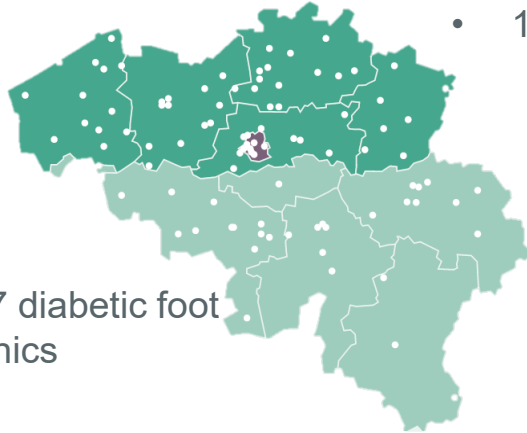
Quality improvement?

Audit – feedback cycles
International guidelines on diabetes care



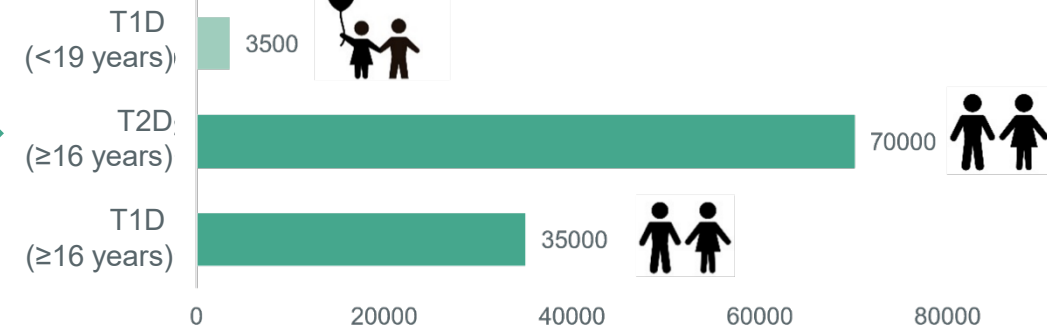
Initiative for Quality Improvement and Epidemiology in Diabetes (IQED)

Hospital based diabetes centers

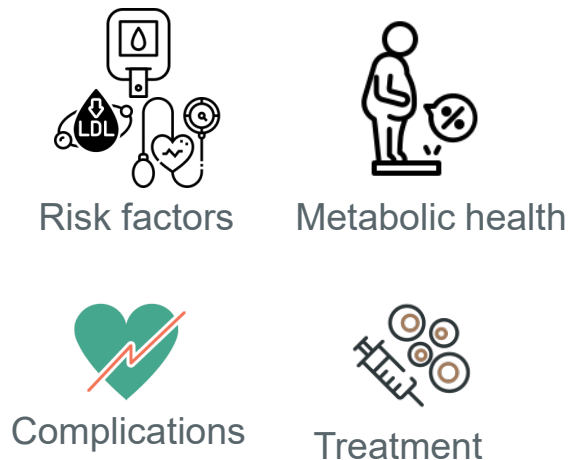
- 101 adult diabetes centers
 - 17 pediatric diabetes centers
 - 37 diabetic foot clinics
- 

Who?

People followed in convention = people with T1D and T2D on intensive insulin treatment

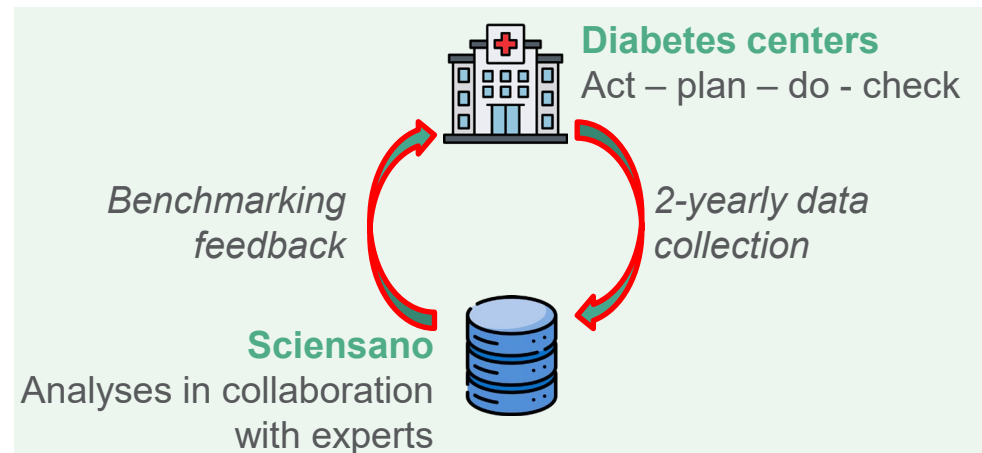


Data? (2-yearly)



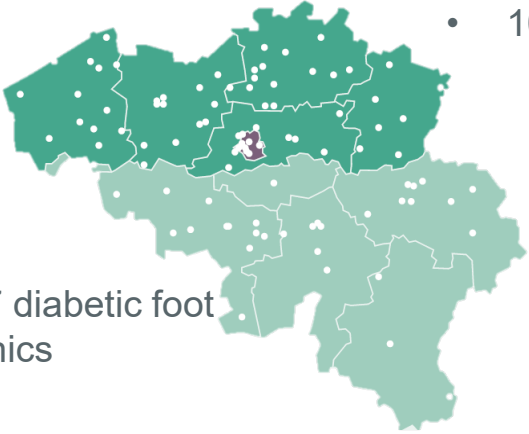
Quality improvement?

Audit – feedback cycles
International guidelines on diabetes care



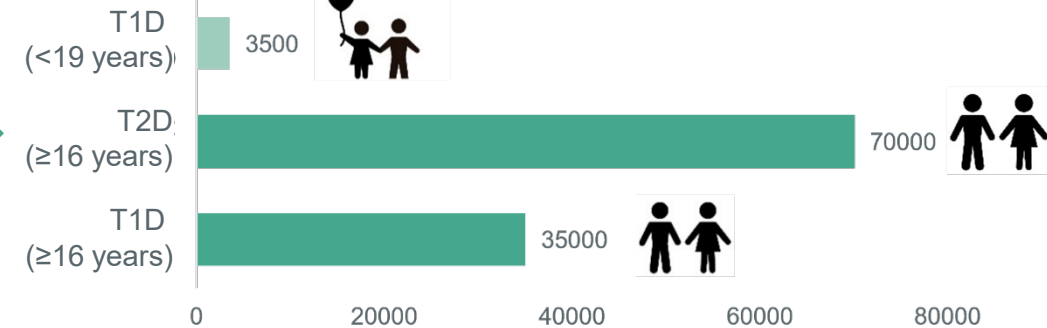
Initiative for Quality Improvement and Epidemiology in Diabetes (IQED)

Hospital based diabetes centers

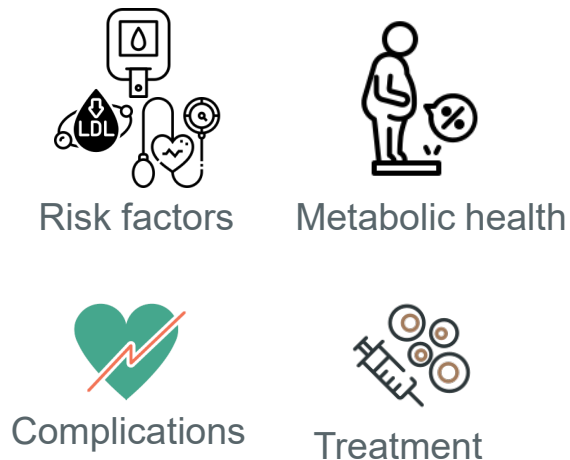
- 101 adult diabetes centers
 - 17 pediatric diabetes centers
 - 37 diabetic foot clinics
- 

Who?

People followed in convention = people with T1D and T2D on intensive insulin treatment

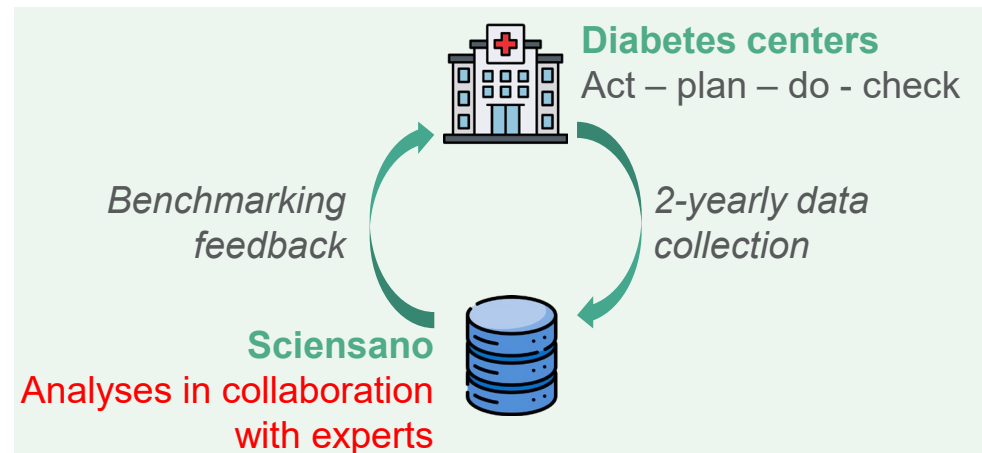


Data? (2-yearly)



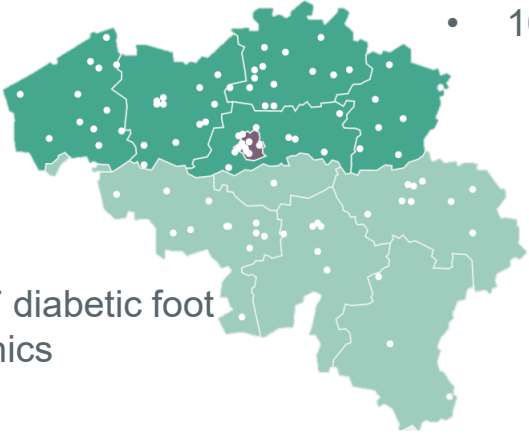
Quality improvement?

Audit – feedback cycles
International guidelines on diabetes care



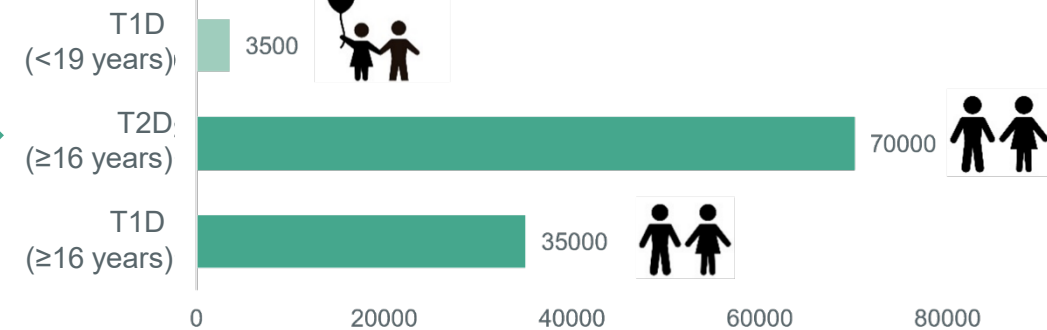
Initiative for Quality Improvement and Epidemiology in Diabetes (IQED)

Hospital based diabetes centers

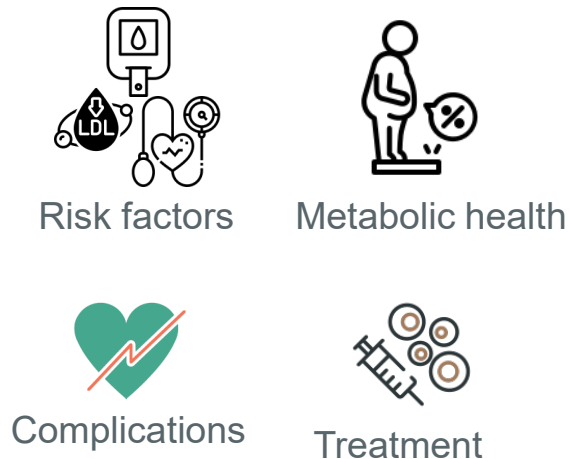
- 101 adult diabetes centers
 - 17 pediatric diabetes centers
 - 37 diabetic foot clinics
- 

Who?

People followed in convention = people with T1D and T2D on intensive insulin treatment

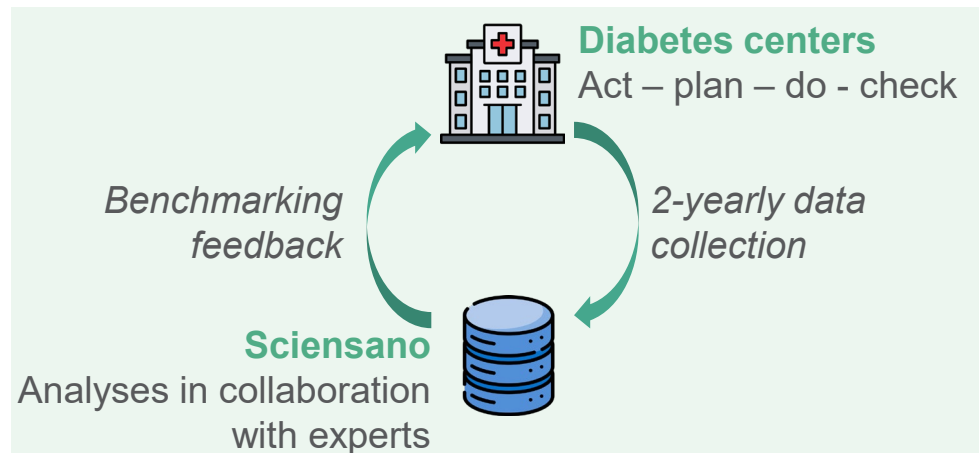


Data? (2-yearly)



Quality improvement?

Audit – feedback cycles
International guidelines on diabetes care

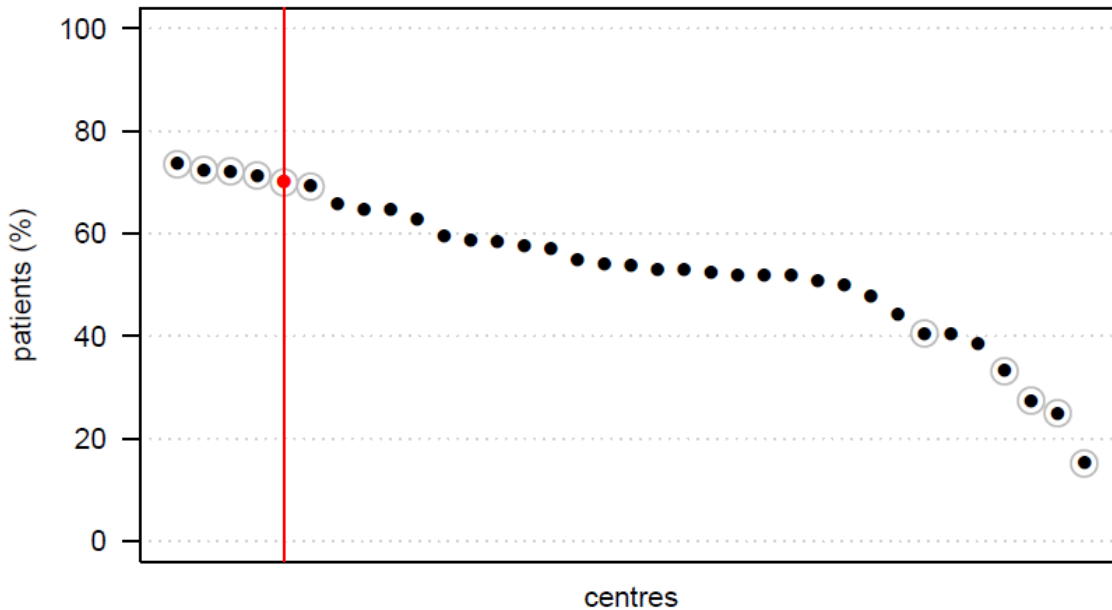


Users and output

- National report for policy makers and care givers
- Flyer for the patients
- Interactive IQED tool
- (inter)national publications

Quality improvement : Audit - feedback cycles

Indicator

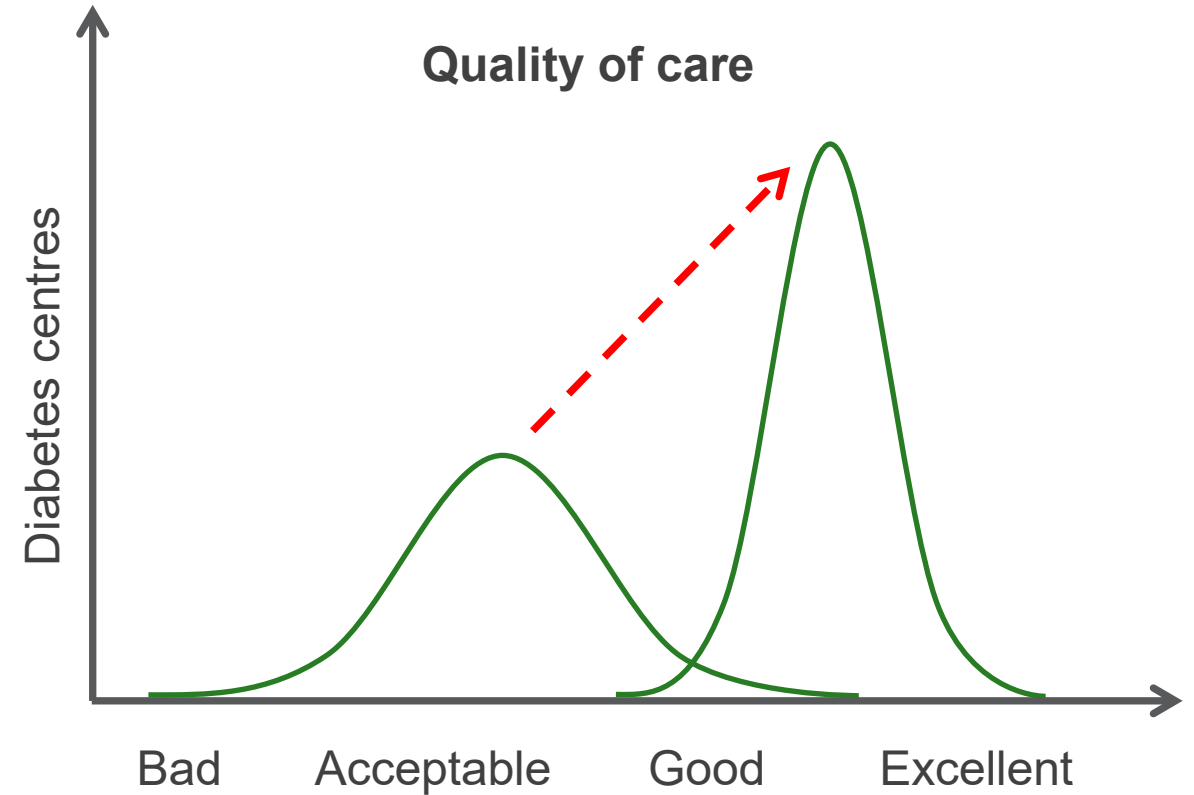


Centre 8 - N = 47 - Votre résultat : 70%

Benchmarking feedback



Quality of care



Stimulate quality of care and reduce variation between centers

IQED tool

 IQED › Initiative for Quality improvement and Epidemiology in Diabetes

Info

General characteristics

Risk factors

HbA1c

Complications

Insulin treatment

Process indicators

Mean outcomes

Cardiovascular therapy

IQED : Initiative for Quality improvement and Epidemiology in Diabetes

Since 1988, Belgian multidisciplinary diabetes centers can enter into an agreement (also called a convention) with the National Institute for Health and Disability Insurance (NIHDI). This convention aims to offer regular medical and educational follow-up as well as materials for self-regulation of glycemia to patients treated with insulin. Since 2001, hospitals with a convention must participate in a quality improvement initiative of data collection with the aim of improving diabetes care. For more info and publications, please visit the [webpage](#).

STUDY POPULATION

IQED is a cross-sectional retrospective observational study, collecting data from individuals with type 1 and type 2 diabetes older than 16 years, followed up in specialized diabetes centers. Data are collected from more than 100 centers across Belgium, covering patient profile, drug treatment, risk factors for developing micro- and macrovascular diabetes complications, and prevalence of acute and chronic complications.

Each center is asked to complete a questionnaire of 10% of their total number of conventioned individuals with diabetes (with a minimum of 25 individuals with type 1 diabetes), using the data listed in their medical records. These are sent to Sciensano in a pseudonymized manner and weighted to the national population (100% sample). IQED assesses care against standards described in high-quality international guidelines (American Diabetes Association (ADA), the European Association for the Study of Diabetes (EASD) and the EASD's bundled guideline with the European Society of Cardiology (ESC)).

The IQED study - as well as the adult diabetes convention - targets adults with type 1 diabetes (T1D) and adults with type 2 diabetes (T2D) with a complex insulin regimen. Patients on continuous subcutaneous insulin infusion (CSII) were among the exclusion criteria for the IQED sample population from 2005 to 2014. Since July 2016, individuals with type 2 diabetes treated with up to 2 insulin injections per day are followed within the "Care Trajectories" (created in 2009) and no longer in the convention.

PARTNERS



CITATION

A. Lavens, R. Ekelson, K. Todorova, Group of experts IQED. Initiative for Quality improvement and Epidemiology in Diabetes. Brussels, Belgium : Sciensano ; 2024



<https://healthinformation.sciensano.be/shiny/IQEDtool/>

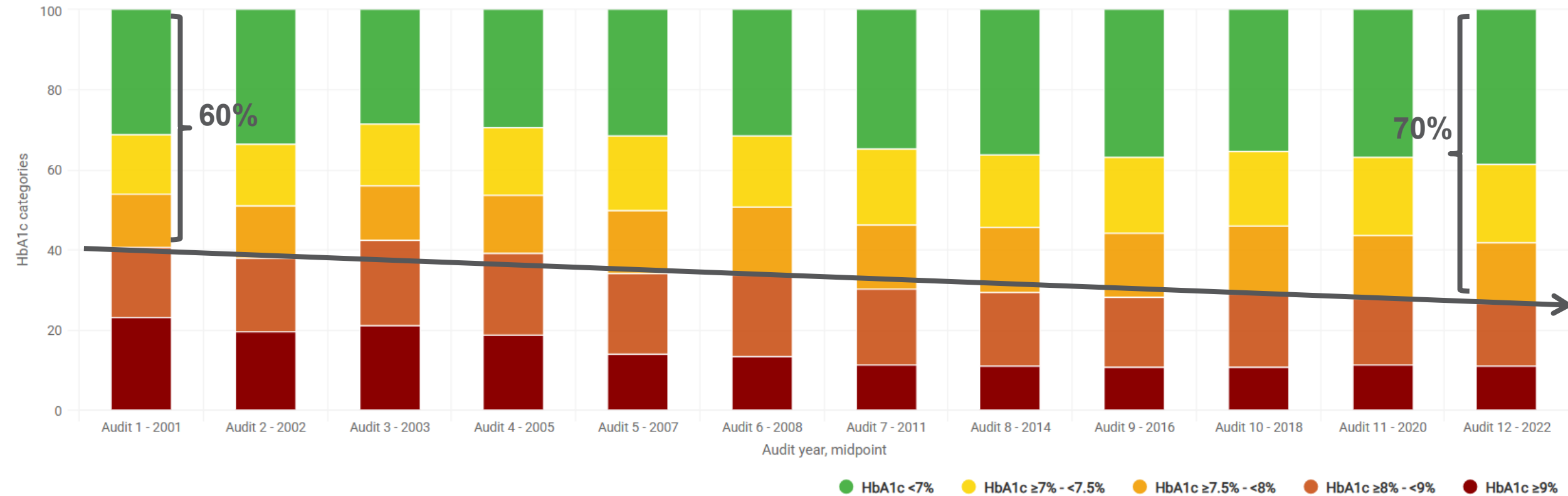
IQED – HbA1c categories – T2D on intensive insulin

Glycemic control: HbA1c categories

(Type 2 diabetes, all ages, both)

- The IQED study is a cross sectional study. Different individuals may have been included in each audit.
- CSII users were excluded from the IQED data collection from 2005 to 2014.
- Since July 2016, individuals with type 2 diabetes treated with up to 2 insulin injections per day are followed within the "Care Trajectories" (created in 2009) and no longer in the convention.
- Values based on less than 10 patients are not shown.

HbA1c <8%
(64 mmol/mol)



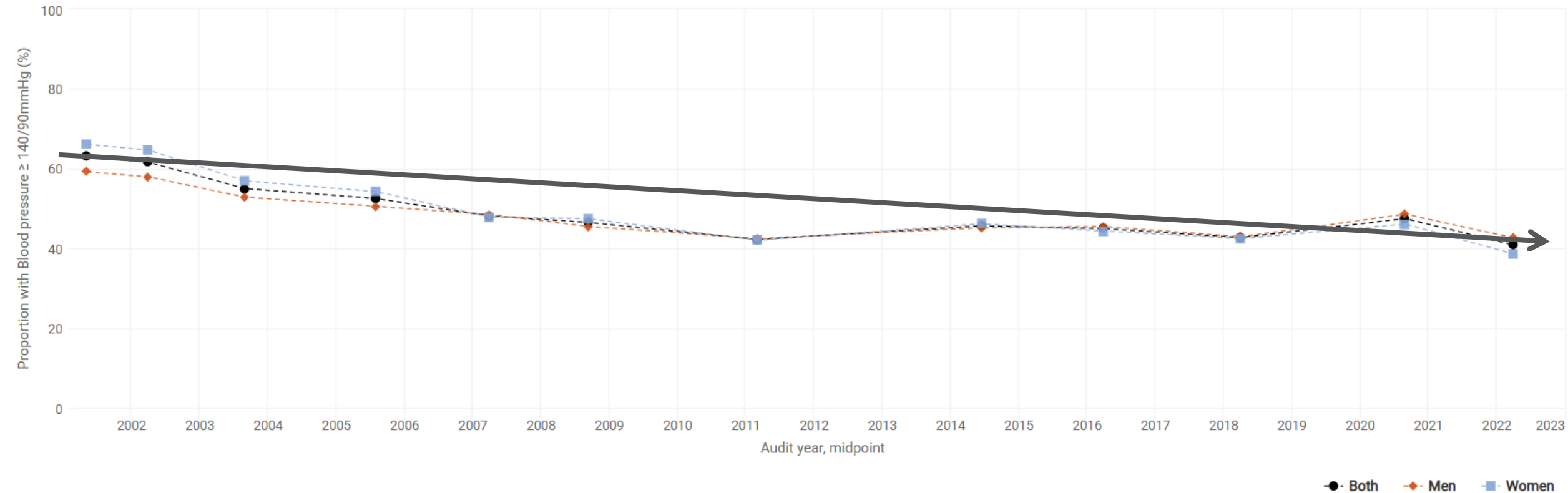
IQED –blood pressure $\geq 140/90$ mmHg – T2D on intensive insulin



Proportion of patients at risk: Blood pressure $\geq 140/90$ mmHg

(Type 2 diabetes, by sex, all ages)

- The IQED study is a cross sectional study. Different individuals may have been included in each audit.
- CSII users were excluded from the IQED data collection from 2005 to 2014.
- Since July 2016, individuals with type 2 diabetes treated with up to 2 insulin injections per day are followed within the "Care Trajectories" (created in 2009) and no longer in the convention.
- Values based on less than 10 patients are not shown.



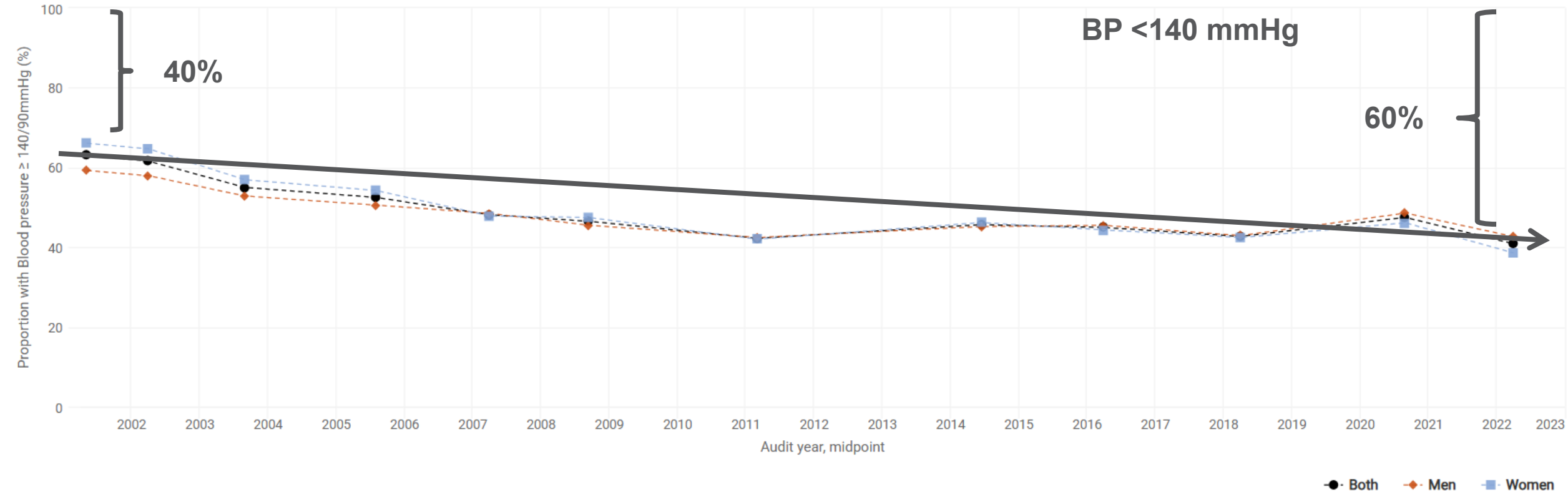
IQED –blood pressure $\geq 140/90$ mmHg – T2D on intensive insulin



Proportion of patients at risk: Blood pressure $\geq 140/90$ mmHg

(Type 2 diabetes, by sex, all ages)

- The IQED study is a cross sectional study. Different individuals may have been included in each audit.
- CSII users were excluded from the IQED data collection from 2005 to 2014.
- Since July 2016, individuals with type 2 diabetes treated with up to 2 insulin injections per day are followed within the "Care Trajectories" (created in 2009) and no longer in the convention.
- Values based on less than 10 patients are not shown.

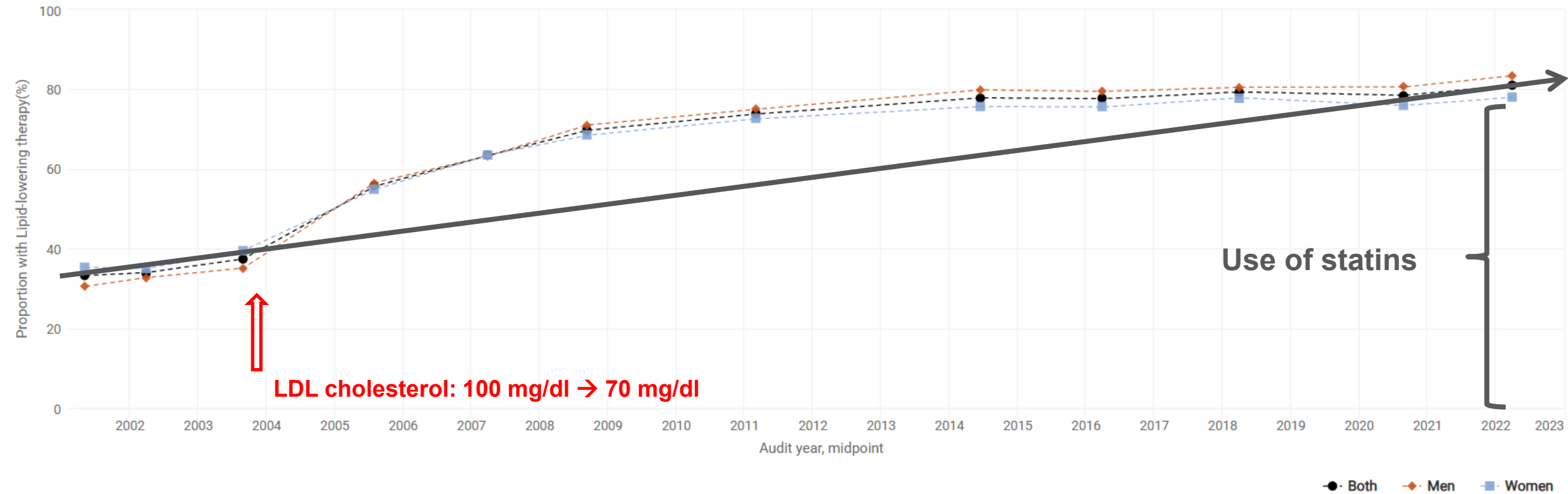


IQED – use of statins – T2D on intensive insulin

Proportion of patients having received (during the audit year): Lipid-lowering therapy

(Type 2 diabetes, by sex, all ages)

- The IQED study is a cross sectional study. Different individuals may have been included in each audit.
- CSII users were excluded from the IQED data collection from 2005 to 2014.
- Since July 2016, individuals with type 2 diabetes treated with up to 2 insulin injections per day are followed within the "Care Trajectories" (created in 2009) and no longer in the convention.
- Values based on less than 10 patients are not shown.



IQED – use of statins – T1D & T2D on intensive insulin

Measuring the WHO global diabetes targets: Belgium – IQED data source

<u>Target</u>	80% of people with diabetes are diagnosed	80% of people with diagnosed diabetes have good control of glycaemia (<8% (64 mmol/mol))	80% of people with diagnosed diabetes have good control of blood pressure (<140/90 mmHg)	60% of people with diabetes of 40 years or older receive statins	100% of people with type 1 diabetes have access to affordable insulin and blood glucose self-monitoring
Measurable		YES	YES	YES	YES
T1D	(BELHES national survey)	69.5%	70.0%	48.7%	100 Insulin and devices are part of the Belgium health benefits package
T2D on intensive insulin	(BELHES national survey)	73.2%	59.1%	81.0%	

Conclusion

- **Four out of five** WHO global diabetes targets **can be assessed using IQED data**.
- Although **all people** living with T1D and T2D requiring insulin do **have access to affordable insulin and blood glucose self-monitoring**, the WHO targets are not all met.
- **Good quality, real-world data is essential** to monitor progress towards defined targets, assess trends and developments, and support evidence-based policy-making.

Contact

Astrid Lavens • Astrid.Lavens@sciensano.be